

2.44 Integrated Production and Separation[☆]

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Glossary

Bioreactor Equipment (reactors) for performing biochemical reaction using biocatalysis such as enzymes and cells.

Continuous operation A way of process operation. In continuous operation, raw materials are continuously fed into a reactor and products are also recovered from the reactor continuously.

Fed-batch A biotechnological batch process which is based on feeding of a growth-limiting nutrient substrate to a culture. The fed-batch strategy is typically used in bio-industrial processes to reach a high cell density in the bioreactor.

Immobilize In bioprocesses, to bind enzymes or cells with a solid or insoluble carrier or to entrap them in a certain space like gel.

Mass transfer Transfer of chemical species between two phases through an interface or diffusion through a phase. The driving force for mass transfer is a difference in concentration. For separation processes, mass transfer determines the rate at which the separation will occur.

2.44.1 Introduction

Bioprocess usually consists of three processes – upstream, production/bioreactor, and downstream processes – as shown in Fig. 1. In most cases, each process is followed by the treatment of various waste materials excreted from the respective process. In the production process, raw materials are converted into products by biochemical reactions employing biocatalyst such as enzymes, microbial cells, animal cells, and plant cells. In the upstream process, materials used for the production process (e.g., raw materials, biocatalyst, and medium) are prepared. In the downstream process, target products are separated from reaction mixture and purified and/or adjusted to a required level.

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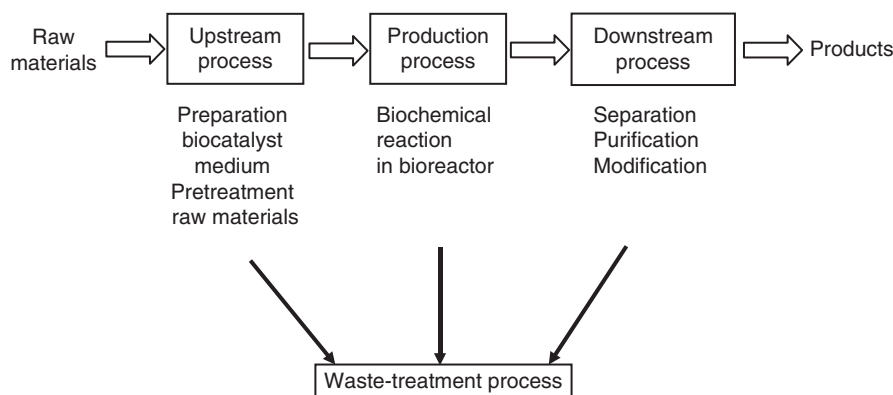


Figure 1 General process flow of bioproduction.

In view of the process economy, the cost for separation and purification process frequently accounts for more than half of the total process cost. Therefore, the downstream process is deeply assessed in order to construct a cost-competitive bioprocess. To attain a required level of purity, many steps for separation and/or purification have been applied sequentially, resulting in an increase in the downstream process cost. Many steps in the process result in not only increases in equipment costs but also decreases in the whole separation efficiency of the process, even though the recovery ratio of each step is high enough. For example, if the recovery ratio of each step is equally α , the whole recovery ratio of the process consisting of n steps is α^n . This implies that the total recovery ratio of five steps decreases to 0.77 even if that of each step (α) is high as 0.95. Therefore, besides improvement of the recovery ratio of each step, a decrease in the number of steps (n) by the process design is much effective for increasing the efficiency of whole downstream process. In the above case, the whole efficiency improves to 0.86 when the steps are decreased from five to three. To decrease the step n , utilization of bioaffinity such as antigen–antibody reaction will be a key technology. For example, the recovery ratio was demonstrated to greatly improve from 16% to 81% by changing a combination process of precipitation, several kinds of chromatography, and electrophoresis into a single step using a monoclonal-antibody column.

Configuration of the downstream process is affected by the features of the upstream and/or production processes. The structure of the downstream process strongly depends on product concentrations from bioreactor. A higher concentration of target product from bioreactor can be treated by a simpler and faster downstream process, resulting in a lower-cost product formation. A remarkable correlation between the product concentrations and the total production costs has been shown. Therefore, bioreactor design and/or operation aiming at attaining higher product concentration and/or decreasing the steps of downstream process are another concern for the whole-process economy. One such approach is the integration of some process steps among upstream, bioreactor, and downstream processes, where chemical and/or biotechnological characteristics of the relevant materials are taken into consideration.

Design of bioreactor itself will be under the concept of process integration. Accumulation of by-products in a bioreactor sometimes inhibits the bioreaction, preventing increased product concentrations, resulting in decreased separation efficiency. The form of biocatalyst, for example, enzyme or whole cells, immobilized or suspended form, also largely affects the efficiency of the downstream process. Immobilized cells or enzymes can be easily removed from the reaction mixture; however, immobilized catalysis frequently decreases reaction efficiency and total reaction rate due to conformational changes of enzymes and the mass transfer resistance. Integration of the downstream process into the bioreactor will bring us other various advantages such as prevention of product inhibition, inhibition of reverse reaction, prevention of product conversion into other chemicals by further reaction, and minimization of product loss due to uncontrolled removal from the system such as evaporation.

As for extracellular products, *in situ* product removal (ISPR), which is the fast removal of product from a producing cell, thereby preventing its subsequent interference with cellular or medium components, has been extensively studied. The extensive reviews by Roffler et al.¹ and Freeman² summarized much of the research on ISPR carried out previously. ISPR primarily requires the identification of properties by which the product differs from the background medium, thus allowing easy removal. On the other hand, intracellular products, which are synthesized and accumulated in cells, require cell disruption, which not only increases the processing steps in the downstream process but also make the processes complex and difficult due to necessity of removing cell lysis materials containing various types of proteins, saccharides, and lipids. In addition, cells themselves are biocatalysts that are produced through cultivation by supplying nutrients including carbon and nitrogen sources. Hence, production of the cells is costly. Disruption of the cells implies discarding precious biocatalyst; it is impossible to realize final products that are less expensive than cells themselves. Therefore, methodologies for the secretion of intracellular products without cell disruption will be one approach for the process integration.

In any process consideration mentioned above, it must always be taken into consideration that most bioproducts such as proteins are sensitive to process conditions such as temperature, pH, and ionic strength, easily losing their activity under inappropriate conditions.

2.44.2 Integration Methodology for Reducing Process Step

The processing pathway of the bioproducts produced using whole microbial cells as biocatalyst from bioreactor is summarized in Fig. 2.

The starting material for the downstream process is either in a liquid phase or in a solid phase (cells) for extracellular or intracellular products, respectively. The latter needs cell separation from the culture broth and product removal from the separated cells by methods such as cell disruption, after which a process similar to that for the former is followed. Process integration aiming at the decrease in processing step can be divided into four categories as shown in Fig. 2. The first is the implementation of the method for removal of bioproducts from culture broth directly by using various biomolecular characteristics, which is indicated as ISPR. The second is an invention of bioreactor itself by means of separation devices such as membrane or cell immobilization. The third is to modify the cell features by biotechnological methods such as genetic modification. The last is the implementation of cell treatment method into the bioreactor by utilizing genetic principles. According to the four categories, strategies for process integration can be discussed to determine which steps in the respective processes can be integrated and/or which steps can be omitted.

In the first category shown in Fig. 2, extracellular products can be directly removed from culture broth containing cells in a bioreactor by one or more methods of ISPR.

In the second category, the separation of cells and supernatant in the culture broth in a bioreactor is considered, where centrifugation is the most popular method. This step can be skipped by cell isolation techniques such as cell-immobilization or membrane separation. The immobilized cells are already in a form of separation from culture liquid. Cell immobilization allows simultaneous product separation during bioreaction from the bioreactor. Membranes can retain cells in a bioreactor during continuous recovery of the culture liquid from the bioreactor.

In the third category, target products are intracellular. One effective methodology for this case is to alter them into extracellular in nature by fusing a secretion sequence to them. As another method, cell surface display is a potential technology for production extracellularly. This method allows performing bioconversion at the outer surface of the cells by displaying enzymes that are responsible for the relevant reaction on the cell surfaces. Thus secreted products or compounds produced extracellularly can be separated from culture broth in the same methods as ISPR.

If it is impossible to use these methods, intracellular products must be separated from cell lysate after cell disruption. This is the fourth category shown in Fig. 2. To skip both the cell separation and cell disruption steps, the cell features are modified genetically

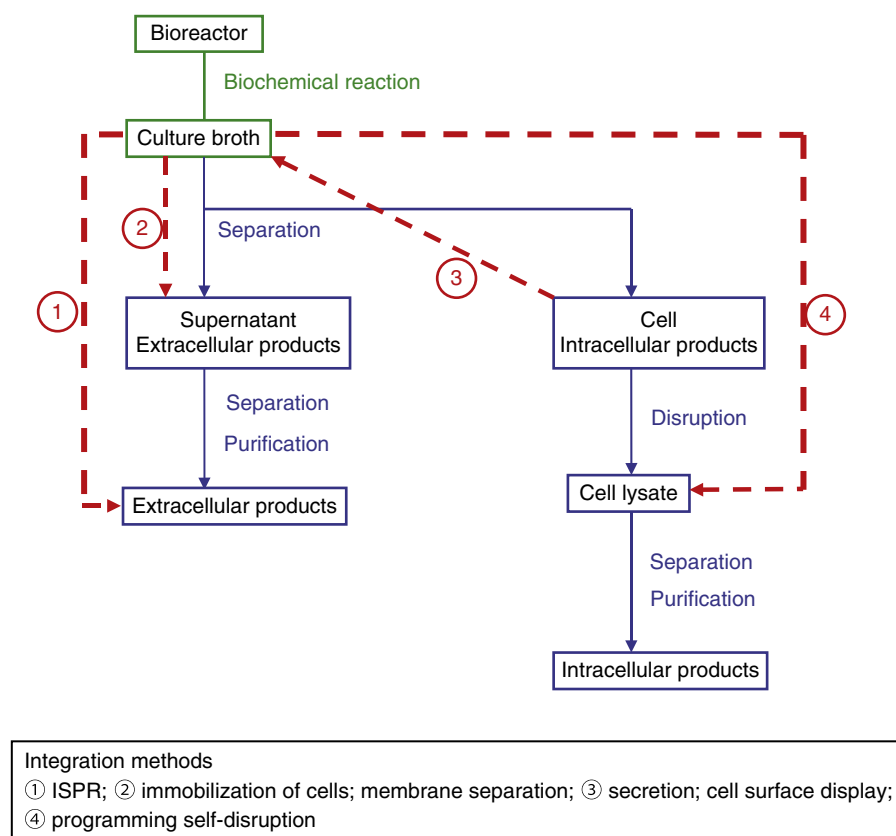


Figure 2 Strategy map for integration of production and downstream processes. Solid lines denote conventional process flow. Broken lines denote the integration process flows which skip the conventional steps.

beforehand for self-disruption under a programmed condition. The condition will be that the programmed cell is auto-lyzed as soon as a sufficient amount of accumulation of the products in a cell is attained. This integration method decreases the number of separation steps and simultaneously minimizes the intracellular products to be degraded or further metabolized into other chemicals.

2.44.3 Cross-sectional Technologies Through Integration Methodology

Integrated production and separation conspicuously demonstrates its efficiency in continuous operation of a bioreactor. Immobilization of biocatalyst and utilization of membrane separation are themselves means of integrated production and separation to skip a cell separation step, but they also have advantages in the continuous operation through the retention of cells in a high concentration in the bioreactor. Continuous operation, immobilization of biocatalyst, and membrane separation are effective cross-sectional technologies through all methods for the integrated production and separation.

2.44.3.1 Continuous Operation

Continuous operation is almost equal to integrated production and separation because it implies continuous bleeding of culture broth, from which products can be separated continuously by an external separation unit. In other words, continuous cell culture lends itself to online integration with the downstream processing. This also allows monitoring of changes as a function of product quality and therefore adjustment of cell conditions as necessary to maintain product consistency.³ Batch processes are difficult to monitor, since the effects of any changes that occur are integrated throughout the entire volume of the vessel. Long-term continuous bioreactions are more economical, and real-time process monitoring is possible, which becomes increasingly important for a number of reasons. For example, mammalian cell culture gives product concentrations of typically 1–100 mg L⁻¹. This high background makes it difficult to monitor product quality in the cell culture medium. Monitoring the purified product from the integrated separation and purification process, instead of the broth, is extremely advantageous.

2.44.3.2 Immobilization of Biocatalyst

Immobilization of enzymes or cells is a widely used method for simplification of the separation process and is employed in various operation modes from batch to continuous operations in a continuous-flow stirred-tank bioreactor (CSTB), fixed-bed plug-flow bioreactor (PFB), or fluidized-bed bioreactor, with and without recycle of the medium. However, one of the greatest merits of immobilized biocatalyst is that it enables continuous or repeated use of precious biocatalyst. Immobilization is frequently coupled with one of the ISPR methods such as evaporation, extraction, and adsorption because it minimizes damage of the biocatalyst from direct contact of the biocatalyst with toxic solvent for extraction or adsorbent in an external unit.²

Enzymes can be immobilized to water-insoluble supports by a wide variety of methods, including physical adsorption, covalent binding or cross-linking, and entrapment. On the other hand, immobilized cells can carry out more complex reactions than immobilized enzymes due to the far more extensive and sophisticated biochemical capabilities of living cells compared with a single enzyme. Whole-cell biocatalysts can also be simpler to prepare. In general, the methods for immobilizing whole cells are similar to those for enzyme immobilization. They are categorized into (1) carrier-free immobilization through cross-linking or flocculation; (2) immobilization onto a preformed carrier through adsorption or covalent attachment; (3) immobilization in the course of carrier formation through entrapment or encapsulation; and (4) immobilization within a preformed semipermeable membrane.

Recently, biofilms, which are accumulated biomasses of microbial cells and extracellular polymeric substances (EPS) on solid surfaces, have attracted attention as a form of natural immobilization^{4,5} or passive immobilization.⁶ Immobilization using biofilms are also classified into three categories, (1), (2), and (4) stated above: (1) granule formed by autoflocculation, (2) biofilms formed onto a carrier surface, and (4) biofilms formed onto one side of a membrane. Conventionally, biofilms have been used in the fields of wastewater and off-gas treatments and bioremediation. Now, they are studied for their application to the production of conventional fermentation products, biofuels, and chemicals.^{7–11} However, the use of biofilms for the production of a chemical has several drawbacks: difficulty in screening a special microbial strain that possesses both high capability to produce the target chemical and the ability to form a biofilm,⁵ a long startup period for biofilm formation,⁵ and difficulty in controlling ideal biofilms with high biocatalytic activity and appropriate thickness in the balance between cell concentration and mass transfer limitation.^{5,9,11}

Very recently, a novel nanofibrous protein AtaA, which mediates notably tenacious adhesion of bacterial cells to various material surfaces ranging from hydrophobic plastics to hydrophilic glass and stainless steel, was discovered.¹² A novel method for bacterial cell immobilization using AtaA was invented. Immobilization with AtaA enabled exclusive cell growth in the support space and only a few cells existed in the bulk medium. Immobilization of resting cells drastically increased cell concentration, depending on the support material; dry cells of approximately 110 g/L could be immobilized onto glass wool.¹³ Immobilized recombinant bacterial cells with AtaA greatly improved the productivity of a chemical through enhancing the tolerance of the cells to a toxic substrate¹⁴ and were efficiently used for repetitive reactions.¹³ This new method is expected to overcome drawbacks of conventional immobilization methods, such as the limitation of mass transfer in the inner part of a gel,^{15,16} fragility of gels, leakage of cells from a support matrix, adverse effect on cell viability and catalytic activity,⁴ and easy detachment from physical adsorption.

2.44.3.3 Membrane Separation

Membrane separation is also combined with other integration methods of production and separation. The principle of the membrane separation is based on the difference in permeability of materials and substances. The driving force for the separation is given by the difference in the pressure, concentration, and electric potential. Fig. 3 shows pressure-driven membrane separation modes depending on the size of separation targets.

Microfiltration (MF) is frequently used for separation of microbial cells from medium in CSTB and PFB in the combination with other ISPR techniques such as product adsorption, extraction, and evaporation. The major mode of application of MF in this context is for retention of microbial cells¹⁷ or mammalian cells and hybridoma cells.³ In addition, like immobilization of cells, cell separation by MF minimizes damage of the biocatalyst from direct contact of the biocatalyst with toxic solvent or adsorbent.² Therefore, it is a growing trend to use MF as an intermediate component for cell separation in a cascade operation rather than the sole means for ISPR. Membranes can also be used for enhanced selective product removal, in which it is essential to maintain a concentration driving force by removing products on the downstream side. For example, membranes have been applied to extractive fermentation, in which organic solvents are used for ISPR, to separate the aqueous and organic phases.

2.44.4 Separation Techniques for the Integration in Terms of Product Characteristics

Separation methods integrated with the production process can be determined by the chemical characteristics of bioproducts. Therefore, products can be categorized for the integration methodology by their chemical characteristics such as volatility, molecular weight, hydrophobicity, charge, solubility, composing chemicals (protein, sugar, lipid, etc.), specific affinity to chemicals, and so on. Possible separation techniques depending on characteristics of products for the integration with production are listed in Table 1.

One of the important and unique points of bioprocesses, compared with conventional chemical processes, is that they can produce protein that can be designed and modified in the cells by genetic manipulation. Proteinaceous products can be previously designed for simplifying the downstream process. For example, introduction of a His-tag at the N-terminus or C-terminus of the protein for simple separation by affinity chromatography is a widely used method in biotechnology. Because the proteinaceous products already contain the separation tool, this method can be considered one of the examples of the integration of production and separation. This method can also be considered to be a way for the integration of the upstream and downstream processes because the cells, which contain the DNA engineered for products integrated with the separation means, are designed before production. Thus, in the case of proteins, this effective method for the integration can be utilized. Introduction of a signal peptide at the N-terminus of target protein for secretion is also categorized in the same methodology for the integration.

When products bind to specific chemicals, separation methods using bioaffinity such as affinity chromatography are some of the most efficient methods for integrated production and separation. Because this method using molecular recognition has high selectivity against a target product, purification of the products can be carried out with high efficiency without additional purification steps. Interactions between antigen and antibody, protein A and antibody (immunoglobulin G (IgG)), and lectin and sugar are famous examples for utilization of affinity separation. Introduction of the His-tag mentioned above is also a method for utilizing bioaffinity.

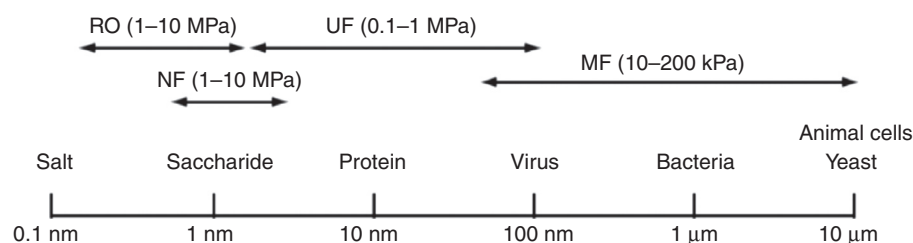


Figure 3 Pressure-driven membrane separation mode depending on the size of separation targets. RO, reverse osmosis; NF, nanofiltration; UF, ultrafiltration; MF, microfiltration.

Table 1 Separation techniques for integration in terms of product characteristics

<i>Characteristics of products</i>	<i>Separation techniques for integration with production</i>
Volatile	Vacuum, pervaporation
Hydrophobic	Organic solvent extraction, hydrophobic adsorption
Hydrophilic	Aqueous biphasic extraction, ion exchange
Low molecular weight	Permeation (dialysis)
Proteinaceous	Secretion, affinity separation
Complexation	Precipitation

2.44.5 Bioreactor Configuration for the Integration of Production and Separation

Operation mode of a bioreactor, that is, batch, fed-batch, or continuous with or without cell retention or immobilization, and specific needs for aeration and sterile operation determine the ease of integration of separation into the production process.² Of course, as mentioned above, whether target products are intracellular or extracellular determines how to design separation modules for subsequent steps after cell separation depending on whether the cell fraction or culture supernatant should be further subjected to purification steps to obtain a final product.

Bioreactor configurations for integration of the separation process can be divided into two types in which the separation is performed within the bioreactor (internally) or outside the bioreactor (externally).² The latter type has a separate loop through which a fraction of the medium is circulated for product separation. Although the internal type has an advantage for rapid equilibration and transfer of the product from each individual cell surface to the product pool, resulting in lower product concentrations in the reactor, which may be critical in the case of chemically reactive or highly toxic products, they also have engineering constraints such as limited solids concentration in stirred-tank bioreactors. Although most membrane modules used to be employed as an external unit, submerged membrane modules assembled by porous hollow-fiber membrane made from polymer materials are frequently employed as an internal unit in a bioreactor.

Extraction with organic solvents is also carried out both internally and externally. In the internal case, a water-immiscible organic solvent is directly added into the bioreactor, and in the external case the medium is circulated through an external extraction unit. In the latter case, by using immobilized cells or membrane, the direct contact between the extracting solvent and the producing cells can be prevented, reducing the toxic effects of the solvent on cell activity. In contrast, extraction by means of an aqueous two-phase system is usually carried out within the bioreactor. Reversible complex formation for ISPR has been carried out with soluble reagents in the medium within the bioreactor.¹⁸ Adsorbents for product immobilization are sometimes directly introduced into a bioreactor.¹ However, external adsorption columns, which are separated from a bioreactor by means of membrane modules¹⁹ or immobilization of either the producing cell²⁰ or the adsorbent²¹ are used when problems arise from the direct contact of the adsorbent with the cells and the medium.

Configurations of bioreactors integrated with separation module are depicted in Fig. 4.

The integrated systems are categorized by the localization of products, extracellular (A to E) or intracellular (F and G) products. Once intercellular products are released into the medium by cell disruption or constrained secretion from cells, same methods with extracellular products can be employed for integration of production and separation. The operation modes are divided into two modes, batch (A, B, C, and F) and continuous modes (D, E, and G). Separation is carried out internally within a bioreactor (A, E, and F) or externally at the outside of the bioreactor (B to D and G). Cell capturing systems utilizing immobilized cells or membrane are also integrated into a bioreactor (C to E). For intracellular products, there is a strategy in which cell disruption system is integrated into the production process (F and G). Because cell disruption in the bioreactor implies the end of biochemical reaction, the process must be a batch system (F). For continuous operation, a fraction of cells should be disrupted externally (G).

2.44.6 Techniques for ISPR

Even intracellular products can be targets of ISPR once they are released into the medium by cell disruption or constrained secretion from cells. Thus, ISPR techniques are the center of the issue in the integration of production and separation. The important ISPR techniques are explained here. Five techniques for ISPR are listed as follows²: (1) evaporation, via vacuum fermentation, gas stripping, or pervaporation, which are effective for low-molecular-weight volatile product such as ethanol and butanol; (2) extraction into another phase, for example, water-immiscible organic solvent, supercritical fluid, or second aqueous phase; (3) size-selective permeation using membranes; (4) reversible complex formation based on chemical reaction with soluble reagents or biological recognition, leading to soluble or insoluble complexes and (5) product immobilization via adsorption or specific binding onto water-insoluble polymeric carriers. Affinity chromatography for the biochemical products is contained in this category.

2.44.6.1 Vacuum and Pervaporation

Volatile fermentation products can be separated during fermentation by maintaining the bioreactor under vacuum so that the products are boiled off at the normal temperature of the fermentation.¹ Ethanol and butanol have been the sole targets of this method. However, a severe problem of this method is the accumulation of toxic nonvolatile byproducts. To solve this problem, continuous vacuum fermentation with liquid bleed has been carried out, where some of the broth from the bioreactor was continuously removed. This method enables feeding a concentrated sugar substrate, resulting in less water requirement in the production process and therefore reduction of wastewater-treatment costs. However, even with liquid bleed for removal of nonvolatile byproducts, large amounts of by-product carbon dioxide must be compressed from the bioreactor pressure up to atmospheric pressure using bulky expensive compressors. To overcome this problem, flash fermentation, in which the bioreactor remains at atmospheric pressure while broth is circulated to a vacuum chamber where products are continuously boiled off, was developed.

In the gas stripping, the fermentation gas is bubbled through the fermentation broth and then passed through a condenser for solvent recovery.²² The stripped gas is then recycled back to the bioreactor and the process continues until all the substrate in the

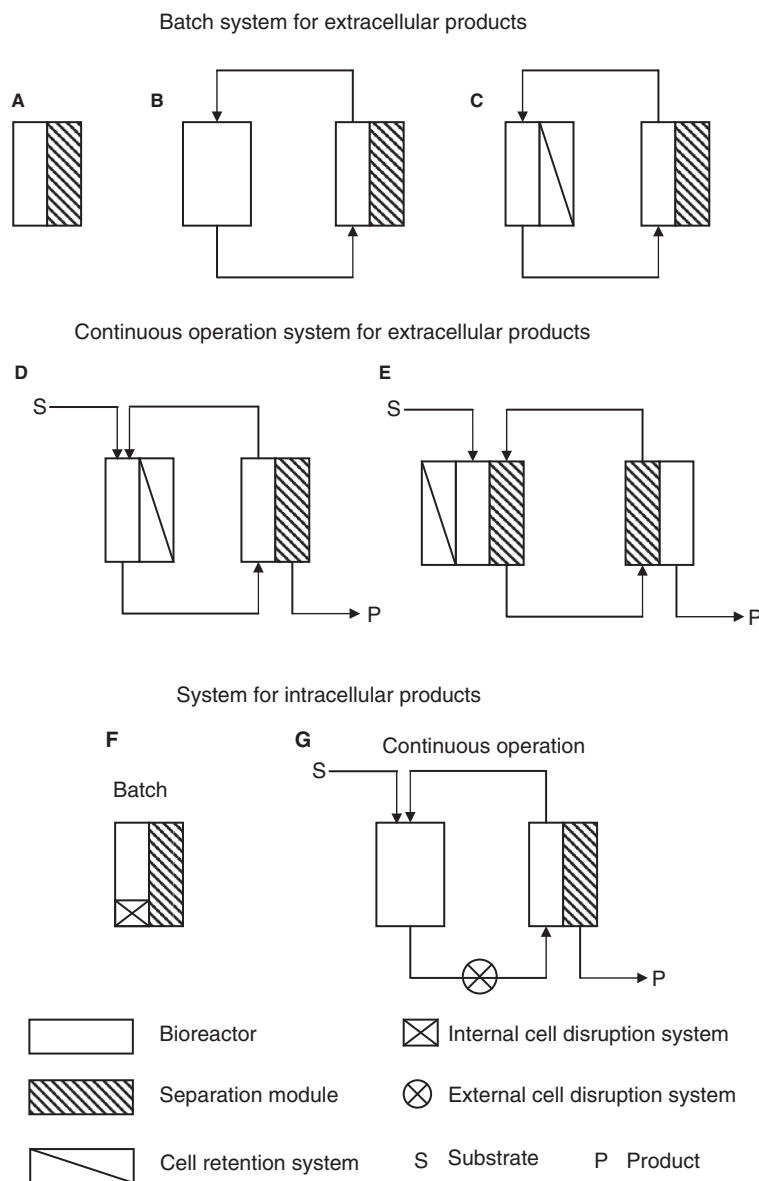


Figure 4 Conceptual schematics for integrated production and separation: (A) batch system integrated with internal separation; (B) batch system integrated with external separation; (C) batch system integrated with internal cell retention system and external separation; (D) continuous operation system integrated with internal cell retention and external separation; (E) continuous operation system integrated with cell retention, internal separation, and regeneration system of the separation carrier; (F) batch system integrated with internal cell disruption; and (G) continuous operation system integrated with external cell disruption and external separation. Modified from Ref. 2.

bioreactor is utilized. Fed-batch fermentation integrated with gas stripping successfully reduced substrate inhibition and increased cell mass in acetone–butanol–ethanol production.²³

Pervaporation is a membrane-based process that allows selective removal of volatile compounds from fermentation broth. The membrane is placed in contact with the fermentation broth, and the volatile liquids or solvents diffuse through the membrane as a vapor which is recovered by condensation.²² Both liquid and solid pervaporation membranes have been used. Pervaporation was identified as the ISPR technique having the greatest potential for improving butanol production.²⁴ Recently, by integration with a pervaporation system using an ionic liquid polydimethylsiloxane ultrafiltration membrane, the overall solvent productivity in continuous acetone–butanol–ethanol fermentation by *Clostridium acetobutylicum* was increased remarkably.²⁵

2.44.6.2 Two-Liquid-Phase Systems

Two-phase liquid systems for integrated production and separation can be divided into the aqueous/organic biphasic system and the aqueous biphasic system. These liquid–liquid extractive bioconversion processes seem to have the greatest potential among the

different approaches to integrating reaction and product recovery.²⁶ In particular, the aqueous/organic biphasic system has been, so far, extensively studied by many researchers and has been at the center of the study about ISPR.

Hydrophobic bioproducts can be extracted during fermentation by contacting the broth with a suitable organic solvent which is insoluble in the broth.¹ Products dissolving into the solvent can later be recovered by distillation or back extraction. Product removal is performed internally or externally, as mentioned in the previous section. The concept of the two-phase partitioning bioreactor can also be applied to controlled delivery of a toxic substrate dissolved in an organic phase to a cell-containing aqueous phase.²⁶ In this case, the solvent should be directly added into the bioreactor (internal removal) to which the toxic substrates are supplied (Fig. 5).

This type of the bioreactor is called two-liquid-phase partitioning bioreactors (TPPBs). It is desirable to choose a solvent that has a high capacity, is selective for the fermentation product, and is relatively nontoxic to the fermenting microorganisms.¹

However, many solvents with a high extraction capacity often have poor selectivity and also exhibit toxicity to the producing cells. In general, a positive correlation is found between hydrophobicity of solvents and nontoxicity for biocatalysts. A measure for hydrophobicity for characterization of solvents is the log *P* value which is the logarithmic of the partition coefficient of a solvent in a water–octanol two-phase system. Solvents with a log *P* value above four are very hydrophobic and generally show no toxic effects on biocatalysts. Organic solvents, for example, dodecanol, dibutyl phthalate, kerosene, dioctyl phthalate, octanol, cyclohexane, silicon oil, dodecane, hexadecane, and tetradecane, have been used in two-liquid phase systems.

According to the solvent selection strategies presented by Bruce and Daugulis,²⁷ it is possible to compose mixtures of solvents that show good extraction capacity as well as good biocompatibility even at relatively high concentrations of toxic solvents. Another problem of the direct addition of solvents to a bioreactor is emulsification. Frequently, the emulsion is so stable because of cell adhesion to oil surfaces that the organic phase containing the products is difficult to be separated. To avoid the emulsification problem, the external extraction in the combination of cell immobilization or use of membrane is preferable to the internal extraction.

Utilization of solvent-tolerant microorganisms is another approach to overcome the problem of the toxicity of solvents added to a bioreactor. Solvent-tolerant bacteria are promising as biocatalysts for the production of highly toxic chemicals in TPPBs. Since Inoue and Horikoshi first (in 1989) reported *Pseudomonas putida* that is capable of growing in culture media containing more than 50% toluene,¹⁹ many bacterial strains such as *Pseudomonas* sp. and *Rhodococcus* sp. that are tolerant to solvents have been isolated. Because in TPPBs, the organic phase acts not only for product removal but also as a toxic substrate reservoir and keeps solvent-tolerant microorganisms in the aqueous phase separated from the substrates and the products, the substrate concentration can be increased with the expectation of the accelerating a chemical conversion rate. In this system, however, mass transfer of substrates dissolved in the solvent (i.e., substrate uptake by microorganisms) is a rate-limiting step in many cases. Employment of microorganisms that take up substrates by direct contact with the organic phase has an advantage in the rapid conversion in TPPBs. Bacterial strains showing monolayer adsorption to oil surfaces have also been reported to be effective for microbial conversion at oil–water interfaces.²⁸

To overcome some of the problems with organic liquids in extractive fermentation, aqueous two-phase systems have been used for the integration of production and separation.¹ Rather than using an organic solvent as the second phase, polymers are added to the broth until two separate phases form. The phases contain 85%–95% water and are normally biocompatible. Microbial cells often remain in one phase while low-molecular-weight products are distributed evenly between phases. Although the product is

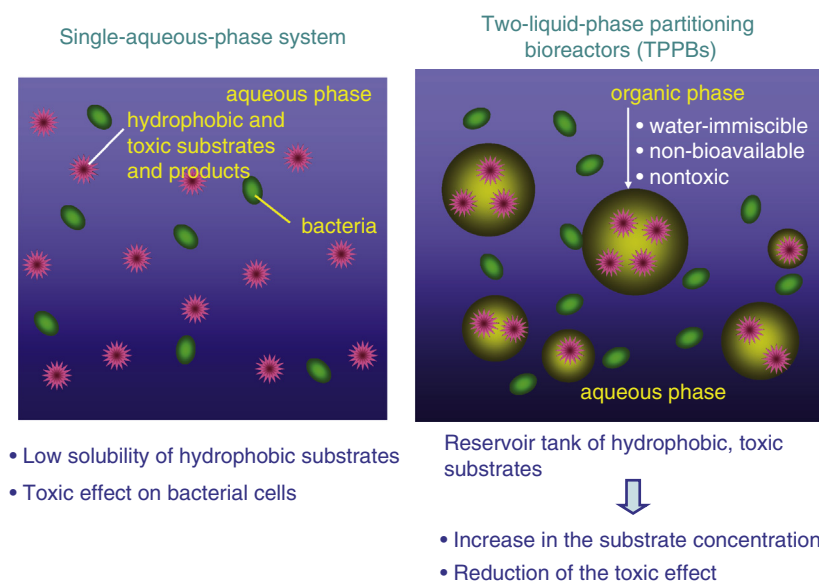


Figure 5 Two-liquid-phase system for internal process integration.

not concentrated by extraction, the phase volumes can be adjusted so that the phase containing the cells is much smaller than the other phase. Most of the product will then be in the phase without cells and can be drawn off and processed by distillation or other means. Polyethylene glycol and dextran are frequently used as polymers to form the incompatible two phases.

2.44.6.3 Size-Selective Permeation

The successful application of membrane modules for separation, up to industrial scale, serves as a sound basis for the integration of membranes with bioreactors.² In dialysis fermentation, a selectively permeable membrane separates a bioreactor from a medium reservoir.¹ Nutrients in the reservoir diffuse to the bioreactor while metabolite products diffuse to the medium reservoir. In general, dialysis fermentation not only relieves product inhibition but also retains cells so that high cell densities are achieved; thus, smaller, less expensive bioreactors could be used. The dialysis fermentation is generally categorized as membrane bioreactor (MBR). The detail was already explained above and also there are many reports about MBRs.

2.44.6.4 Complex Formation

Reversibly soluble polyelectrolytes, whose solubility in aqueous solutions is dependent on factors, such as pH, ionic strength, and temperature, are used in the integrated system as supports for enzymes or affinity ligands.¹⁸ Their dual nature takes advantage of the soluble form during enzymatic reactions and the insoluble form during separation of products.

The ability cyclodextrins to form crystalline-insoluble complexes with organics can be used for a selective separation of dilute products in fermentation.²⁹ It was demonstrated that butanol was selectively precipitated from the acetone-butanol-ethanol and butanol-iso-propanol systems and butyric acid from the butyric acid-acetic system. Employment of epichlorohydrin-cross-linked β -cyclodextrin in ISPR for product immobilization also showed high yield and easy pure product recovery in γ -decalactone production from castor oil by a filamentous fungus.³⁰

2.44.6.5 Product Adsorption

Product removal by immobilization via adsorption onto polymeric matrices has been demonstrated for a relatively large variety of products such as ethanol, salicylic acid, cycloheximide, anthraquinones, alkaloids, monoterpenes, and tissue plasminogen activator.² Activated carbon, zeolites, polystyrene beads, cellulose, and ion exchanges served as the adsorbing matrix, mainly on the basis of hydrophobic interactions. A limitation of these materials may be the nonspecific adsorption of organic compounds from solution. This results in the removal of both products and substrates leading to decreased product removal efficiencies as well as increased substrate loadings to compensate for sorbed materials. However, hydrophobic polymeric resins can be effective for specific adhesion of hydrophobic products such as butanol from culture broth containing hydrophilic substrates.³¹ Thermoplastic polymers can be an alternative to both immiscible organic solvents and solid-phase adsorbents.³² Uptake of small molecules into thermoplastic polymers is analogous to that of uptake into organic solvents, which operate on an adsorption mechanism, while solid adsorbents operate on a surface adsorption phenomenon. These polymers have been successfully used partition and deliver toxic organic molecules in biphasic systems. For example, a polyether ester thermoplastic that consists of cylindrical-shaped pellets is used to recover a toxic product, 3-methylcatechol, from the cell broth.

2.44.7 Process Integration by Biotechnology

Biotechnological methods are powerful tools for the integration of production and separation processes. However, the methods are a pool of various ideas and still in the state-of-the-art meaning a bit far from systematic description of them. Here, some examples of such biotechnology applicable to the integration are discussed.

2.44.7.1 Secretion of Intracellular Proteins

Secreted production of proteins is one of the most important strategies for integrated production and separation of proteinaceous products, in particular, recombinant proteins. Intracellular production requires many steps in the downstream process from cell disruption, separation of inclusion body, solubilization, and refolding of a target protein product before purification. For the secretion of intracellular proteins, secretion pathways have been utilized. In bacteria, the general secretory pathway called Sec system is well known.³³ The Sec system consists of membrane proteins SecY, SecE, and SecG, which form a channel on a cytoplasmic membrane for translocation of unfolding proteins, and the intracellular protein SecA, which mediates to conveyance of pre-secretory proteins into the channel depending on energy from adenosine triphosphate (ATP) hydrolysis. The pre-secretory protein possesses a signal peptide consisting of 20–40 amino acids at N-terminal region as a precursor. The pre-secretory protein passes through the channel pore in an unfolding extended string form.^{34,35} By this system, protein is secreted out of cells in the case of Gram-positive bacteria, or into the periplasm in the case of Gram-negative bacteria. Therefore, addition of a signal peptide sequence to the N-terminal of a target protein enables the secreted production of the protein that is originally an intracellular protein. Expression vectors designed for this purpose are commercially available. Although the Sec system has been studied for efficient

protein-secreted production system for more than 20 years, there remains a limitation; some proteins that are important in industry cannot be secreted effectively by the Sec system.

Recently, the secondary general secretory pathway, twin-arginine translocation pathway (Tat pathway), first found in a chloroplast of plant cells has been shown to be conserved in bacteria.^{36,37} Proteins secreted by this pathway also possess a signal peptide, which contains a specific motif of Arg-Arg- Φ - Φ (Φ denotes a hydrophobic amino acid) at its N-terminal. Unlike the Sec system, the Tat pathway transports folded proteins using proton motive force. In the Tat pathway, there is a hitchhiker mechanism, in which an enzyme complex is formed in the cells, and one of the constitutive subunits that are appended to the Tat signal peptide co-translocates the other subunits in the complex.³⁷ These unique properties of the Tat pathway are expected to expand the possibility of application in a protein bioprocess. This system has been applied to the secreted production of Fab antibody fragment in *E. coli*.³⁸ Generally, disulfide bonds are not formed in the cytoplasm due to its reducing environment. The heavy chain fragment tagged with the Tat signal peptide and the light chain fragment were produced, and the disulfide bonds were formed in the cytoplasm of the mutant *Escherichia coli* strain, whose cytoplasm is in oxidative environment. By this strategy, correctly folded Fab fragment was successfully secreted.

2.44.7.2 Utilization of Affinity Separation for Proteinaceous Products

Proteins such as antibodies and enzymes are typical and important products from bioprocesses. Proteinaceous products are unstable and easily degraded by proteases released from cells. Integration of production and product separation minimizes the loss of proteinaceous products by disintegration. Generally speaking, proteins physiologically function through the specific interaction with other molecules. Affinity separation utilizes these interactions, that is, affinity between respective proteins and their ligands such as antigen-antibody, enzyme-substrate, enzyme-inhibitor, cytokine-receptor, and hormone-receptor. There have been many commercially available affinity separation systems for proteinaceous products (Table 2), and they can also be employed in the integrated production and separation by utilizing inline affinity chromatography.

After all, affinity chromatography has much higher purification efficiency and higher recovery than other types of chromatography and also is capable of treating a large volume of samples so that it should be used in an early stage of the purification steps. Therefore, affinity chromatography is one of the efficient methods for integrated purification into production for proteinaceous products. For example, restriction endonucleases were produced as a model protein by continuous two-stage cultivation with a recombinant *E. coli* strain, which was integrated with continuous cell disintegration and purification by affinity chromatography.³⁹ In this study, the enzymes were expressed as fusion proteins consisting of protein A from *Staphylococcus aureus*, and Fc-region of IgG-sepharose-FF column was used for the affinity chromatography.

2.44.7.3 Programming of Self-disruption

As mentioned previously, the intracellular products have to be processed first by cell disruption after harvesting cells from broth in bioreactor, then transferred to the succeeding separation processes. On the contrary, extracellular products are unnecessary to harvest cells and their disintegration. If the products are modified to be excreted genetically as mentioned above, the process for cell treatment can be skipped. However, such modification is not necessarily effective for all kinds of products and cells, especially for nonproteinous products. In the case of difficulty in excretion modification, another idea comes out. One such idea is to genetically modify the cell features into self-disruptive after sufficient amount of product is accumulated in the cell. Fig. 6 shows an

Table 2 Examples of commercially available carrier for affinity chromatography

Carrier	Target for separation
Nickel	His-tag-fused protein
Glutathione	Glutathione S-transferase (GST)-fused protein
Protein A	IgG
IgG	Protein A-fused protein
Heparin	Fibronectin, fibroblast growth factor, hepatocyte growth factor
Concanavalin A	Glycoprotein, membrane protein
Lectins	Glycoprotein
Streptavidin	Biotinylated protein
Calmodulin	ATPase, protein kinase, phosphodiesterase
Gelatin	Gelatinase, fibronectin
Cibacron blue 3 GA	NAD(P)-dependent protein, interferon
Procion Red HE-3B	NADP-dependent protein, matrix metalloproteinases, tissue inhibitor of metalloproteinases
Lysine	Plasminogen, plasminogen activator
Arginine	Prothrombin, plasminogen activator
Benzamidine	Serine protease

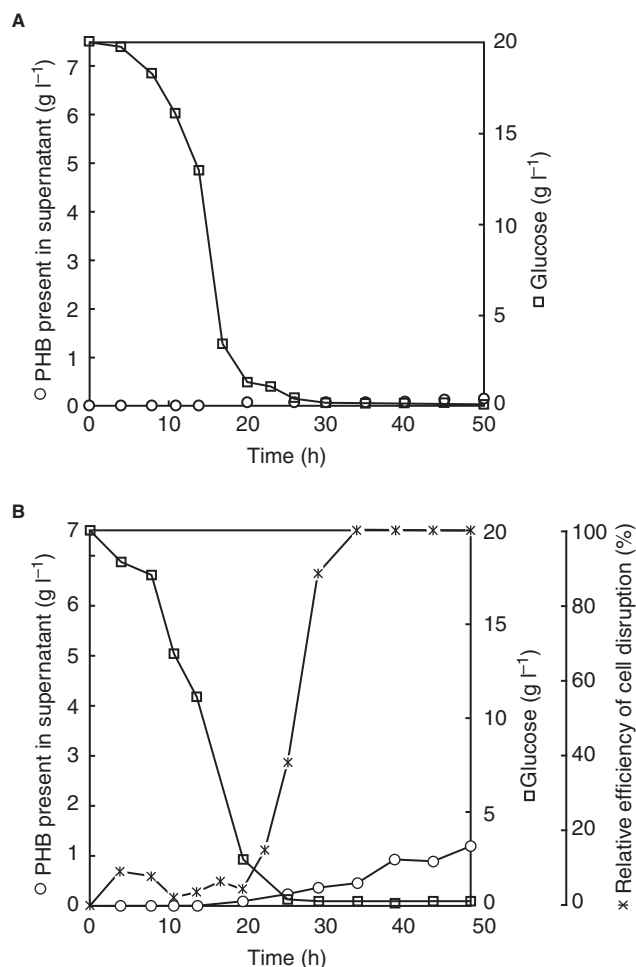


Figure 6 Programmed self cell-disruption: (A) wild-type strain and (B) transformant. Cell disruption of transformant was initiated after complete exhaustion of glucose in medium followed by release of PHB accumulated in the cell (B), while wild-type strain showed scarce release of PHB due to no cell disruption (A). Modified from Hori K, Kaneko M, Tanji Y, Xing X-H, Unno H. 2002. Construction of self-disruptive *Bacillus megaterium* in response to substrate exhaustion for polyhydroxybutyrate production. *App. Microbiol. Biotechnol.* 59: 211–216.

example for such modification, where product, polyhydroxybutyrate (PHB), synthesized and accumulated in *Bacillus megaterium* intracellularly was successfully solubilized into the supernatant by a program in which lysis gene from *Bacillus amyloliquefaciens* phage was engineered into the cell expressing the lytic function by triggered by the exhaustion of glucose in the medium.⁴⁰

Strategic viewpoints on cell disruption in bioprocess design, partially aiming at the process integration, are discussed in the review article by B. Balasundaram et al.⁴¹

2.44.8 Perspective for the Process Integration

Bioprocesses are now expected as alternative production processes to the chemical production process developed so far because they are considered to be energy-saving processes under mild conditions (ambient temperature and pressure), decrease emission of pollutants, employ carbon neutral raw materials, utilize organic waste as raw materials, and so on. However, many bioprocesses encompass the drawbacks of high production costs. To make it possible to adopt the bioprocess as an alternative process to conventional one, process integration will surely be one strategy. Research on integration so far is mostly based on specific ideas. Although most of them are a combination of conventional separation tools of hardware, soft approaches are also employed through biotechnological modification of cells like excretion of products and adhesiveness for immobilization given by a tenacious adhesin,^{12–14} modification of product characteristics appropriate for processing, effective utilization of raw materials and biocatalysts by simultaneous production of effective products aiming at the decrease of production cost.⁴² Approaches to process integration by biotechnological features will surely be accelerated in a variety of integration types by the development of new technologies.

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